

EXPERIMENT – 9

CODE:

```
// Program Sketch
```

```
int trig = 10;
```

```
int echo = 9;
```

```
long duration;
```

```
int cm;
```

```
void setup()
```

```
{
```

```
  pinMode(trig, OUTPUT);
```

```
  pinMode(echo, INPUT);
```

```
  Serial.begin(9600);
```

```
}
```

```
void loop()
```

```
{
```

```
  // Send 10  $\mu$ s pulse to trigger
```

```
  digitalWrite(trig, LOW);
```

```
  delayMicroseconds(2);
```

```
  digitalWrite(trig, HIGH);
```

```
  delayMicroseconds(10);
```

```
  digitalWrite(trig, LOW);
```

```
  // Read echo pulse
```

```
  duration = pulseIn(echo, HIGH);
```

```
// Calculate distance in cm
```

```
cm = duration * 0.034 / 2;
```

```
// Display distance
```

```
Serial.print("Distance = ");
```

```
Serial.print(cm);
```

```
Serial.println(" cm");
```

```
delay(500);
```

```
}
```

OUTPUT:

